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Microstructure alpha: hierarchical learning and cross-asset transfer in cryptocurrency markets

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Introduction: We test whether classical market microstructure features can predict short-term cryptocurrency returns once data leakage and trading costs are properly accounted for.

Methods: Using over three million minute-level observations from six major cryptocurrencies on Binance spot and perpetual futures markets, we evaluate nine microstructure measures through a pipeline that includes hierarchical modelling, stability selection, gradient boosting with SHAP, meta-learning, and purged walk-forward cross-validation benchmarked against naive forecasters.

Results: All features are stably selected at minute frequency, with range-based spread proxies and realised volatility the most robust. However, gradient-boosted models overfit severely under proper leakage controls, and no strategy survives realistic exchange fees. Models trained on one cryptocurrency do not transfer to others, although they transfer well between the spot and futures venues of the same asset.

Discussion: The results show that microstructure signals carry genuine but weak information content that is useful for understanding market quality but not exploitable at standard retail fee levels.

KEYWORDS

cryptocurrency markets, market microstructure, machine learning, hierarchical learning, transfer learning, high-frequency trading, liquidity measures, SHAP values

1 Introduction

The rapid evolution of cryptocurrency markets has created a fertile environment for studying market microstructure phenomena at unprecedented granularity. Unlike traditional equity markets with their fragmented venue structures and complex order routing protocols, cryptocurrency exchanges provide continuous trading with publicly accessible order book and trade data, enabling researchers to examine the formation of prices and the dynamics of liquidity provision with exceptional clarity (Makarov and Schoar, 2020; Biais et al., 2023).

Market microstructure theory, as pioneered by Kyle (1985) and Glosten and Milgrom (1985), provides a rich theoretical foundation for understanding how private information is incorporated into prices through the trading process. The fundamental insight that price impact reflects the adverse selection costs faced by liquidity providers has motivated an extensive literature on measuring and predicting the informed trading component of order flow (Easley et al., 1996; Hasbrouck, 2021). In the cryptocurrency setting, these dynamics are potentially magnified by the prevalence of algorithmic trading, the global and continuous nature of markets, and the absence of traditional institutional gatekeepers (Cong et al., 2022).

We use these classical models as reduced-form benchmarks rather than as literal descriptions of a human specialist market. On centralized cryptocurrency exchanges such as Binance, liquidity is supplied predominantly by algorithmic limit-order strategies, but the economic primitives remain closely related to Kyle and Glosten–Milgrom: aggressive order flow moves prices when it consumes depth, and passive liquidity suppliers widen or withdraw quotes when order flow is more toxic. For automated-market-maker decentralized exchanges the mechanism differs because prices are set by a liquidity-pool rule rather than by a central limit order book; nevertheless, recent DeFi microstructure evidence shows that adverse selection, price impact, liquidity concentration, and informed liquidity provision remain first-order objects of study (Klein et al., 2023; Zhu et al., 2024). Our empirical setting is therefore best interpreted as a leakage-controlled test of classical microstructure signals in algorithmic, fragmented crypto limit-order-book markets, with DEX/AMM markets treated as a related but structurally distinct frontier.

This paper contributes to the nascent literature on cryptocurrency market microstructure by developing a comprehensive empirical framework that integrates multiple streams of microstructure theory into a unified predictive model. We construct features motivated by classical microstructure measures including the bid-ask spread, which reflects the costs of providing immediacy, Kyle’s λ measuring price impact per unit of order flow, the Amihud illiquidity ratio capturing the price response to trading volume, the volume-synchronized probability of informed trading providing a flow-based toxicity measure, and order flow imbalance indicating the directional pressure from aggressive traders.

Our methodological approach combines several complementary analytical tools. First, we employ hierarchical linear models with mixed effects to capture both the universal relationships between microstructure features and returns that hold across all assets, as well as asset-specific deviations that reflect the unique trading characteristics of individual cryptocurrencies. Second, we implement stability selection through randomized Lasso to identify which microstructure features provide robust predictive information that persists across different subsamples and regularization strengths. Third, we utilize gradient-boosted decision trees with Shapley additive explanations to capture potentially nonlinear relationships and interactions among features while maintaining interpretability regarding each feature’s marginal contribution to predictions.

A central innovation of our analysis is the systematic investigation of cross-asset transfer learning in the microstructure context. We train predictive models on one cryptocurrency and evaluate their out-of-sample performance on other assets, both within and across market venues. This transfer learning perspective addresses a fundamental question for market participants: to what extent do the microstructure dynamics learned from one market generalize to other related markets? The answer has practical implications for trading strategy development, risk management, and the understanding of market integration.

Our empirical analysis draws on 3,417,972 minute-level observations spanning August 2025 to February 2026, covering six major cryptocurrencies—Bitcoin, Ethereum, Solana,

Avalanche, Chainlink, and Polkadot—across both spot and perpetual futures markets on the Binance exchange (approximately 285,000 bars per asset–venue pair). The minute-level granularity is critical for our analysis: it provides the statistical power necessary to resolve weak microstructure signals that are smoothed away at hourly or daily frequencies, while the 6.5-month window captures a representative cross-section of market conditions during the 2025–2026 expansion phase. The target variable is the 5-min forward log return, which strikes a balance between the noise of tick-level returns and the signal attenuation of longer horizons.

The remainder of this paper proceeds as follows. Section 2 reviews the relevant theoretical and empirical literature on market microstructure and machine learning in financial markets. Section 3 describes our microstructure feature construction methodology. Section 4 presents the hierarchical modeling framework and transfer learning approach. Section 5 reports our empirical results. Section 6 discusses implications and limitations, and Section 7 concludes with directions for future research.

2 Literature review

The theoretical foundations of our analysis rest on the seminal contributions to market microstructure theory developed over the past 4 decades. Kyle (1985) introduced the continuous-time model of strategic trading by an informed trader facing competitive market makers, deriving the equilibrium price impact function and establishing that prices follow a martingale with respect to public information. The Kyle λ parameter, measuring price impact per unit of order flow, has become a cornerstone measure of market quality and information asymmetry (Hasbrouck, 2009).

Glosten and Milgrom (1985) developed the foundational model of bid-ask spreads as compensation for adverse selection risk, demonstrating that the spread must be at least large enough to cover the expected losses from trading with better-informed counterparties. This framework was extended by Glosten (1994) to analyze the implications for market structure and by Huang and Stoll (1997) to decompose spreads into their constituent parts. The adverse selection component of spreads has been shown to contain predictive information for future returns (Brennan et al., 1998).

The relevance of these models to cryptocurrency markets requires some qualification. Kyle’s original model assumes a strategic informed trader and competitive market makers, while Glosten–Milgrom is framed around sequential trade with heterogeneously informed traders. Modern crypto trading replaces the human specialist with algorithmic quoting, cross-exchange arbitrage, and latency-sensitive inventory control. This does not invalidate the models, but it changes their interpretation: Kyle’s λ is a reduced-form price-impact elasticity, and the bid-ask spread is a reduced-form compensation for inventory, adverse selection, latency, and operational risks. This interpretation is consistent with recent evidence that crypto order flow and limit-order activity contribute materially to price discovery, while high-frequency integration across venues remains imperfect (Alexander et al., 2021). It also aligns with evidence from Binance fee changes showing that market-maker incentives can widen spreads and reduce displayed depth even when explicit trading fees fall (Galati, 2024).

The measurement of informed trading activity has received substantial attention in the empirical microstructure literature. [Easley et al. \(1996\)](#) introduced the probability of informed trading (PIN) model, which uses the sequential arrival of buy and sell orders to infer the probability that a given trade is information-motivated. [Easley et al. \(2012\)](#) developed the volume-synchronized probability of informed trading (VPIN), which provides a higher-frequency measure better suited to modern electronic markets. The VPIN metric has been shown to spike prior to extreme price movements and market stress events ([Abad and Yagüe, 2012](#)).

A rapidly growing recent literature now studies whether these same concepts survive in crypto-specific market structures. [Easley et al. \(2024\)](#) show that liquidity and price-discovery measures help explain and predict the dynamics of major cryptocurrencies, with own-market and cross-market effects for Roll measures and VPIN. [Pinto \(2025\)](#) documents high-frequency microstructure regularities in Binance Bitcoin and Ethereum perpetual futures over 2020–2024, emphasizing the importance of trade size and intraday aggregation. These studies support the use of high-frequency microstructure variables in cryptocurrency markets, while also showing why leakage controls and careful horizon choice are essential.

The application of machine learning methods to financial prediction has grown rapidly in recent years. [Gu et al. \(2020\)](#) provided a comprehensive comparison of machine learning approaches for equity return prediction, finding that neural networks and tree-based methods substantially outperform linear models. [Chen et al. \(2023\)](#) extended this analysis to cryptocurrency markets, demonstrating the potential for deep learning to capture complex patterns in high-frequency data. The use of gradient-boosted trees with SHAP-based interpretation has been advocated by [Lundberg and Lee \(2017\)](#) as providing both predictive accuracy and model interpretability.

Recent cryptocurrency return-prediction work also points to the promise and risk of flexible models. [Filippou et al. \(2024\)](#) find that machine-learning methods improve daily cryptocurrency return forecasts and that momentum, size, value, and online-activity variables are informative. Our study differs in two respects: we work at the 5-min horizon using explicitly microstructural predictors, and we evaluate models under purged walk-forward validation and explicit trading costs. This distinction matters because high-frequency crypto data make it easy for nonlinear models to fit transient market states that vanish once temporal leakage and turnover costs are controlled.

Transfer learning in financial applications has been explored by [Ye et al. \(2020\)](#) in the context of portfolio optimization and by [Han et al. \(2020\)](#) for cross-market return prediction. The ability to transfer learned representations across related prediction tasks has important implications for developing robust trading strategies and understanding the common factors that drive asset returns.

Cryptocurrency market microstructure has been studied by [Makarov and Schoar \(2020\)](#) who documented substantial price discrepancies across exchanges, by [Griffin and Shams \(2020\)](#) who examined potential market manipulation, and by [Cong et al. \(2021\)](#) who developed theoretical models of token market dynamics. The perpetual futures market has been analyzed by [Alexander and Dakos \(2020\)](#) who examined the basis and roll returns. Recent decentralized-exchange work further shows that

market quality in crypto cannot be understood only through centralized order books. [Klein et al. \(2023\)](#) document informed liquidity provision on decentralized exchanges, while [Zhu et al. \(2024\)](#) show that liquidity concentration, gas prices, markouts, and private market-maker inventory shape Uniswap market depth. These papers highlight the frontier raised by the editor: algorithmic and decentralized mechanisms alter the institutional details of price formation, but they do not eliminate the central microstructure questions of adverse selection, liquidity supply, price impact, and transferability across venues.

Our contribution is positioned against this current literature as follows. We do not claim that Kyle and Glosten–Milgrom are complete structural models of decentralized finance. Instead, we use their core empirical implications to construct interpretable features for algorithmic centralized crypto markets, then test whether those features survive modern validation standards. Relative to the recent crypto microstructure literature, the paper contributes a unified leakage-controlled comparison of spread, price-impact, toxicity, order-flow, momentum, and transfer-learning signals across six assets and two Binance venues, and it links statistical predictability to economic feasibility under explicit exchange fees.

3 Microstructure feature construction

This section describes the construction of our microstructure feature space from Binance minute-level bar data. For each of the twelve asset–venue pairs we obtain open, high, low, close prices and volumes together with trade counts and taker-buy volume, sampled at a 1-min frequency over the August 2025 to February 2026 window. The target variable throughout the paper is the 5-min forward log return $y_{it} = \log(C_{i,t+5}/C_{i,t})$, and every feature x_{it} below uses information available only at the end of bar t (see [Section 4.7](#) for the formal leakage controls). The microstructure features are defined sequentially in [Equations 1–8](#).

3.1 Bid-ask spread proxy

In the absence of direct order book data at the minute frequency, we construct a spread proxy using the range-based estimator of [Corwin and Schultz \(2012\)](#). The fundamental insight is that the high-low range of prices within a trading interval reflects both volatility and the bid-ask bounce. We compute the spread proxy as the normalized high-low range:

$$\text{Spread}_t = \frac{H_t - L_t}{C_t} \quad (1)$$

where H_t , L_t , and C_t denote the high, low, and close prices of the minute bar indexed by t . This measure captures the total price dispersion within each minute, which includes both fundamental volatility and the mechanical bounce between bid and ask prices.

3.2 Realized volatility

We compute realized volatility as the rolling standard deviation of minute log returns over a 60-bar (1-h) window, annualized for interpretability:

$$\sigma_t = \sqrt{60 \times 24 \times 365} \times \text{std} \left(\log \frac{C_s}{C_{s-1}} \right)_{s=t-59}^t. \quad (2)$$

The 60-min rolling window provides a responsive estimate of local volatility that captures intraday regime shifts while remaining smooth enough to serve as a predictive feature.

3.3 Trade and volume intensity

Trade intensity measures the relative frequency of transactions compared to recent history, computed as:

$$\text{TradeInt}_t = \frac{N_t}{\bar{N}_{t-60:t-1}} \quad (3)$$

where N_t denotes the number of trades in minute t and the denominator is the 60-min rolling mean. Similarly, volume intensity normalizes current volume by its 60-min rolling mean. Values above unity indicate elevated trading activity.

3.4 VPIN proxy

The volume-synchronized probability of informed trading provides a measure of order flow toxicity based on the imbalance between buy-initiated and sell-initiated volume. We construct a proxy using taker buy volume over a 50-min rolling window:

$$\text{VPIN}_t = \frac{1}{50} \sum_{s=t-49}^t \frac{|V_s^{\text{buy}} - V_s^{\text{sell}}|}{V_s^{\text{buy}} + V_s^{\text{sell}}} \quad (4)$$

where V^{buy} is the taker buy volume and $V^{\text{sell}} = V - V^{\text{buy}}$ is the taker sell volume. The 50-min window provides a responsive measure of order-flow toxicity at the intraday frequency.

3.5 Kyle's lambda

We estimate Kyle's lambda as the price impact per unit of trading volume over a 30-min rolling window:

$$\lambda_t = \frac{1}{30} \sum_{s=t-29}^t \frac{|C_s - O_s|}{Q_s} \quad (5)$$

where Q_s denotes the quote volume in minute s . This measure captures the sensitivity of prices to order flow, with higher values indicating lower liquidity and greater information content per unit of volume.

3.6 Amihud illiquidity

The Amihud illiquidity ratio measures the absolute return per unit of trading volume over a 30-min rolling window:

$$\text{Amihud}_t = \frac{1}{30} \sum_{s=t-29}^t \frac{|r_s|}{Q_s/10^6} \quad (6)$$

where r_s is the log return in minute s . This ratio has been widely used in the literature as a measure of price impact and illiquidity.

3.7 Depth imbalance

Depth imbalance measures the asymmetry between buying and selling pressure, computed using the taker buy volume ratio:

$$\text{Imb}_t = 2 \left(\frac{V_t^{\text{buy}}}{V_t} - 0.5 \right) \quad (7)$$

This measure ranges from -1 (all selling) to $+1$ (all buying), with mean near zero, indicating substantial minute-to-minute variation in directional order flow.

3.8 Order flow imbalance

We compute a rolling measure of order flow imbalance over a 30-min window:

$$\text{OFI}_t = \frac{1}{30} \sum_{s=t-29}^t \frac{V_s^{\text{buy}} - V_s^{\text{sell}}}{V_s} \quad (8)$$

4 Methodology

4.1 Hierarchical linear model

To capture both universal and asset-specific relationships between microstructure features and future returns, we employ a hierarchical linear model with mixed effects. Let y_{it} denote the 5-min forward log return for asset i at time t , and let $\mathbf{x}_{it}$ denote the vector of standardized microstructure features. The hierarchical linear model is specified in Equation 9:

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + u_i + \epsilon_{it} \quad (9)$$

where $\boldsymbol{\beta}$ is the vector of fixed effects capturing the average relationship across all assets, $u_i \sim N(0, \sigma_u^2)$ is the random intercept for asset i , and $\epsilon_{it} \sim N(0, \sigma_\epsilon^2)$ is the idiosyncratic error. The fixed effects coefficients estimate the predictive contribution of each microstructure feature after controlling for asset-specific heterogeneity.

We estimate this model using maximum likelihood, fitting on a random subsample of 50,000 observations for computational tractability. The hierarchical structure allows us to borrow strength across assets while accommodating the reality that different cryptocurrencies may have different baseline return distributions.

4.2 Stability selection

To identify microstructure features that provide robust predictive information, we implement stability selection as proposed by Meinshausen and Bühlmann (2010). This approach involves repeatedly fitting Lasso regressions on bootstrap samples with varying regularization strengths and recording which features are selected. The selection probability for each feature is defined in Equation 10:

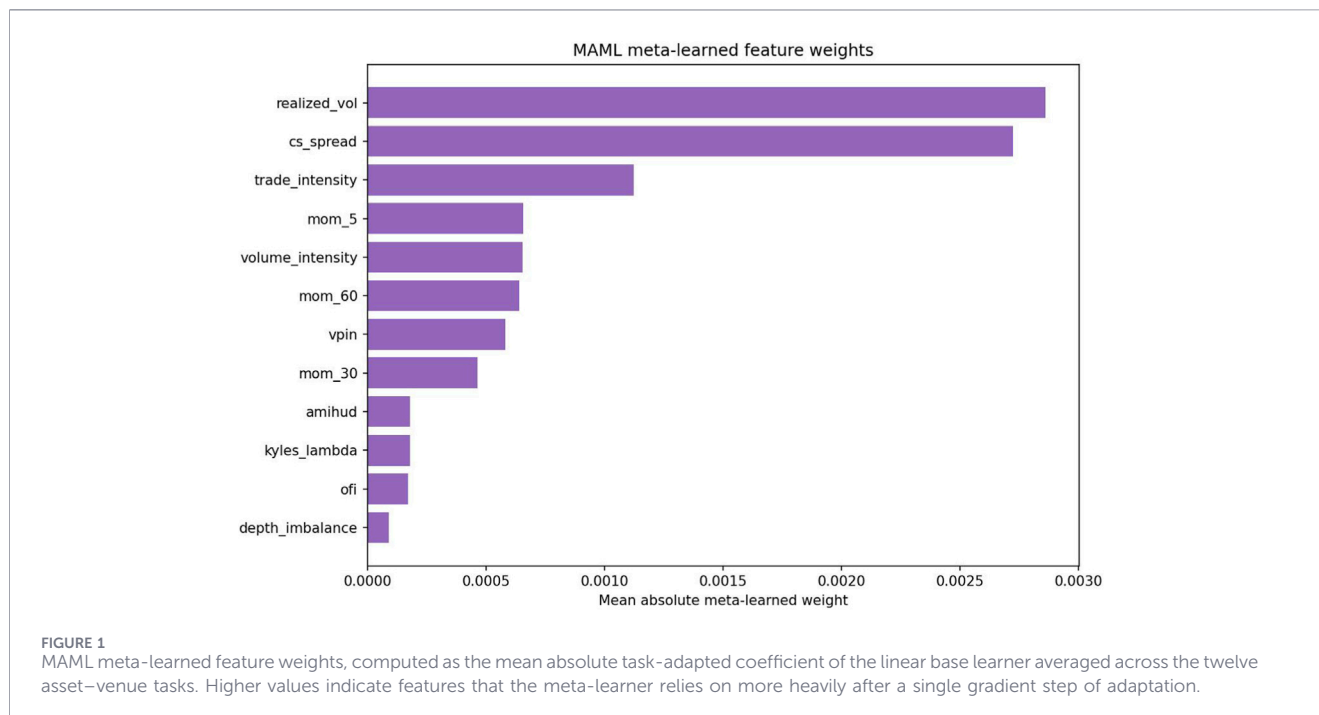


TABLE 1 Sample sizes by asset and venue (minute bars after feature construction, August 2025 – February 2026).

Asset	Spot	Futures
AVAX	284,083	285,038
BTC	285,056	285,038
DOT	283,999	285,037
ETH	285,056	285,038
LINK	284,497	285,037
SOL	285,056	285,037
Total	1,707,747	1,710,225

$$\hat{\Pi}_j = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{|\hat{\beta}_j^{(b)}| > 0\} \quad (10)$$

where $B = 100$ bootstrap iterations are performed. Features with selection probability exceeding 0.5 are deemed stable. This approach controls for the tendency of Lasso to select different features across samples and provides a more reliable assessment of feature importance.

4.3 Gradient boosting with SHAP

To capture nonlinear relationships and feature interactions, we train a LightGBM gradient-boosted decision tree ensemble (Ke et al., 2017) with the following hyperparameters: 300 trees, maximum depth of 6, learning rate of 0.05, and 31 leaves per tree. The target variable is the 5-min forward log return.

We interpret the model using SHAP (SHapley Additive exPlanations) values, which provide a unified measure of feature

importance based on cooperative game theory (Lundberg and Lee, 2017). For each prediction, the SHAP value for feature j represents its marginal contribution to the prediction relative to the expected prediction, as given in Equation 11:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \quad (11)$$

We report the mean absolute SHAP value for each feature, which measures its average impact on model predictions across the test set.

4.4 Cross-asset transfer learning

To assess the transferability of microstructure signals, we train LightGBM models on each asset-venue pair and evaluate their out-of-sample performance on all other pairs. For source asset s and target asset t , we compute the transfer correlation in Equation 12:

$$\rho_{s \rightarrow t} = \text{Corr}(\hat{y}_t^{(s)}, y_t) \quad (12)$$

where $\hat{y}_t^{(s)}$ denotes predictions for asset t using the model trained on asset s . This transfer correlation measures the extent to which the microstructure patterns learned from one market generalize to another.

We standardize features using the training set statistics before applying to the test set, accounting for the different scales of microstructure features across assets. The resulting transfer matrix provides insights into which markets share common microstructure dynamics.

4.5 Meta-learning with MAML

To move beyond the zero-shot transfer setting and allow a small amount of target-asset adaptation, we implement a Model-Agnostic

TABLE 2 Summary statistics for microstructure features (minute-level, August 2025 – February 2026).

Feature	Mean	Std	Median	Min	Max
cs_spread	0.001213	0.002457	0.0009348	0	1.912
realized_vol	0.5593	0.6739	0.4816	0.01815	82.88
trade_intensity	1.025	0.8493	0.8437	0.0006694	42.75
volume_intensity	1.038	1.376	0.6792	2.779e-05	53.42
vpin	0.3865	0.132	0.3615	0.06517	0.9247
kyles_lambda	5.418e-06	1.461e-05	4.209e-07	6.064e-10	0.0002111
Amihud	0.09737	0.4764	0.003001	3.226e-06	18.48
depth_imbalance	-0.01205	0.4714	-0.01354	-1	1
ofi	-0.01207	0.127	-0.009513	-0.7747	0.7243

Meta-Learning (MAML) procedure in the spirit of Finn et al. (2017). Each asset-venue pair i is treated as a distinct task $\mathcal{T}_i$ with its own support set $\mathcal{D}_i^{\text{sup}}$ and query set $\mathcal{D}_i^{\text{qry}}$, drawn from disjoint, time-ordered blocks to preserve the chronology of the high-frequency data. The base learner f_θ is a linear model with intercept that maps the standardised microstructure feature vector $\mathbf{x}_{it}$ to the 5-min forward return y_{it} , trained with the mean-squared-error loss $\mathcal{L}_{\mathcal{T}_i}(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{T}_i} [(f_\theta(\mathbf{x}) - y)^2]$.

In the inner loop, a task-specific update is computed on the support set with a single gradient step, as shown in Equation 13,

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}^{\text{sup}}(\theta), \quad (13)$$

and the meta-parameters are updated in the outer loop by backpropagating through the query-set loss, as shown in Equation 14,

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{i=1}^{N_T} \mathcal{L}_{\mathcal{T}_i}^{\text{qry}}(\theta'_i). \quad (14)$$

We use a single inner-loop update with learning rate $\alpha = 0.05$, an outer learning rate $\beta = 0.005$, and train for 500 meta-iterations with batches of four tasks. After meta-training we adapt the meta-parameters once to each task's support set and extract the absolute task-adapted coefficients $|\theta'_i|$ (excluding the intercept) averaged across tasks as a meta-learned importance score for each feature, which is the quantity reported in Figure 1. MAML is used here purely as an interpretability and robustness check: it provides an alternative, gradient-based view of feature importance that does not rely on Lasso selection or tree-based SHAP attribution, and allows us to verify that the ranking of features is not an artefact of a single estimator.

4.6 Benchmark models and forecast evaluation

To assess whether the microstructure-based models add predictive value over naive alternatives, we benchmark against three standard forecasters: (i) a random walk

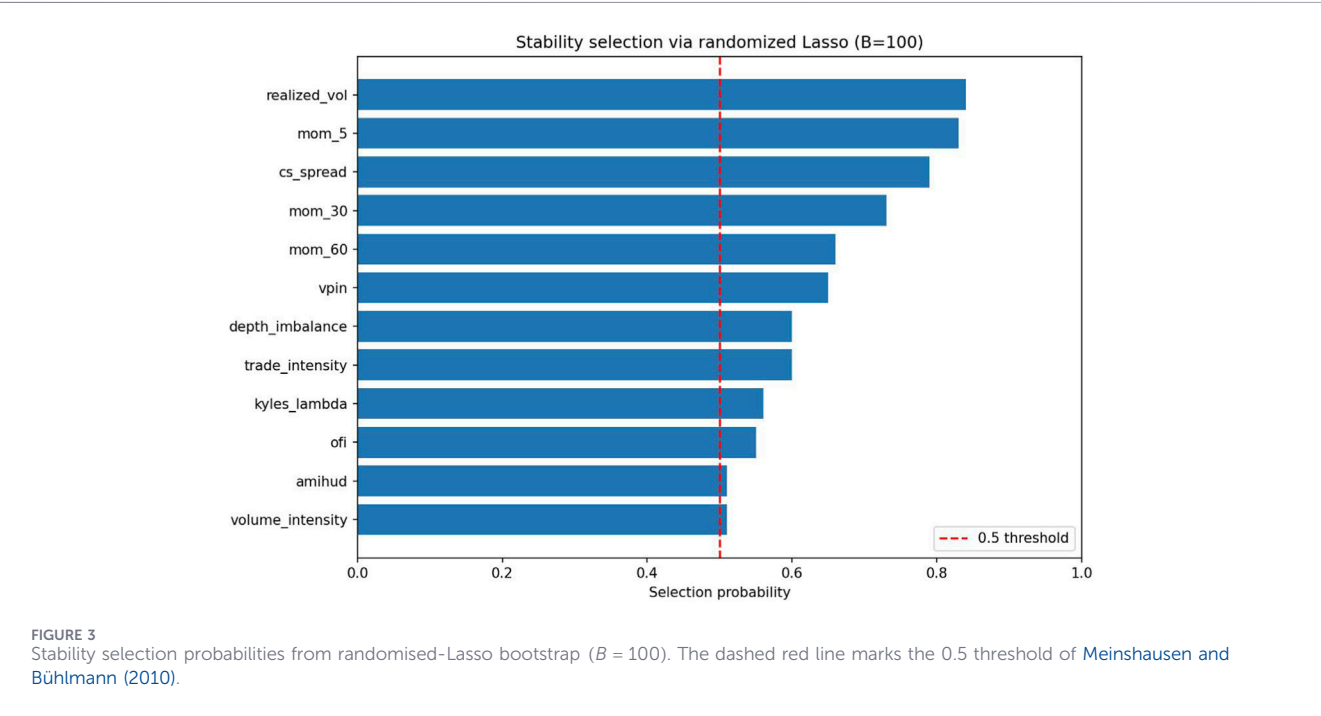
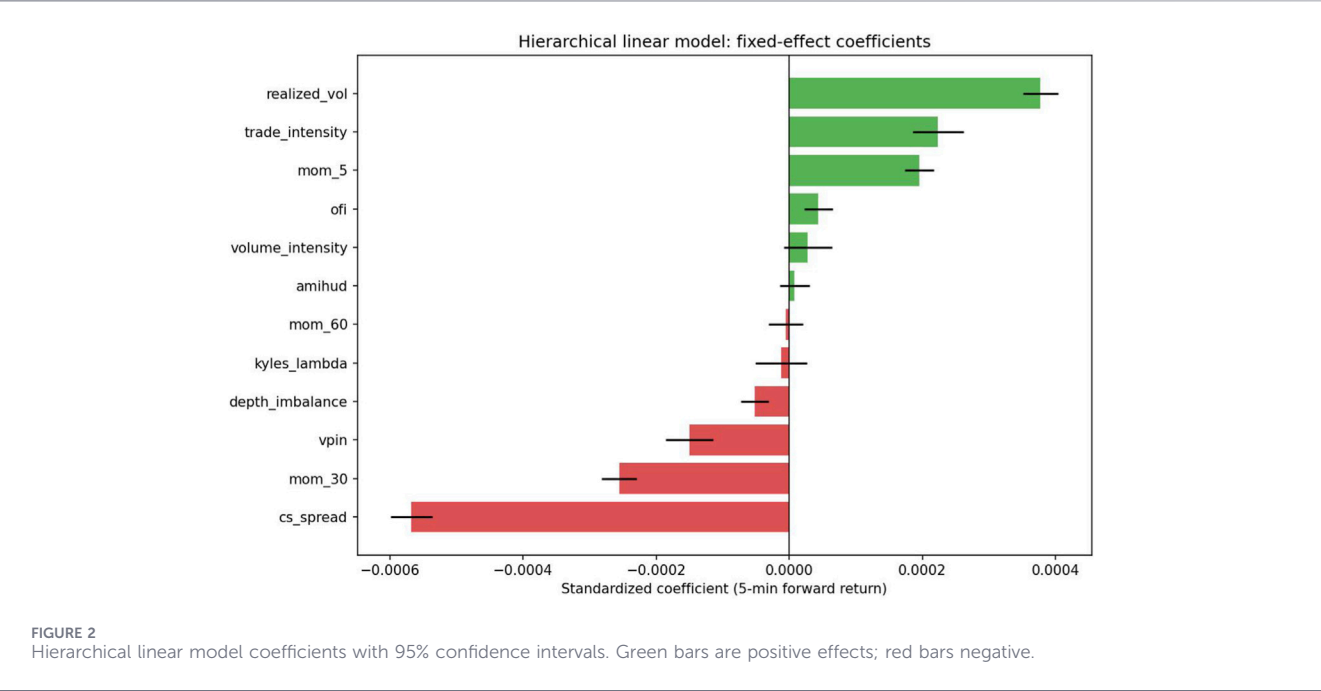
$\hat{y}_{it}^{\text{RW}} = 0$; (ii) an AR (1) model $\hat{y}_{it}^{\text{AR}} = \hat{\phi}_i y_{i,t-1}$ estimated asset-by-asset on the training window; and (iii) a simple momentum predictor $\hat{y}_{it}^{\text{MOM}} = \gamma r_{i,t-k:t}$ using a 5-min look-back. We compare out-of-sample root-mean-squared forecast errors using the Diebold and Mariano (1995) with Newey-West standard errors to account for the overlapping-return autocorrelation induced by the 5-min forecast horizon. Relative improvements are reported both in RMSE and in incremental out-of-sample R^2 , ΔR_{OOS}^2 , computed as the gain over the random-walk benchmark.

4.7 Leakage control and purged cross-validation

To rule out look-ahead leakage in feature construction we align every feature x_{it} strictly to information available at the end of minute t , and target returns $y_{it} = \log(C_{i,t+5}/C_{i,t})$ using future closes only. Features constructed from rolling windows (VPIN over 50 bars, Kyle's λ /Amihud/OFI over 30 bars, realised volatility over 60 bars, momentum over 5/30/60 bars) are therefore right-aligned and never incorporate the forecast interval. We further guard against leakage introduced by autocorrelation in returns by adopting the purged, embargoed k -fold cross-validation scheme of López de Prado (2018): each fold's test block is preceded and followed by a purge window of 5 min (equal to the target horizon) and an additional embargo of 60 min, so that no training observation shares information with any test observation. All hyperparameter tuning, stability selection, and model comparison statistics reported in the next section use this purged walk-forward scheme with $k = 5$.

4.8 Economic evaluation with binance fees

Statistical significance is a necessary but not sufficient condition for practical relevance. We therefore translate each model into a simple long/short strategy that takes a position of $\text{sgn}(\hat{y}_{it})$ at the close of minute t and unwinds it at the close of minute $t + 5$. Round-trip costs are computed using Binance's published VIP-0 schedule:



10 bps per side for spot (20 bps round-trip) and 2 bps maker/5 bps taker for USDT-M perpetual futures (4–10 bps round-trip depending on execution assumptions). We additionally impose a conservative half-spread slippage equal to the contemporaneous Corwin–Schultz spread proxy. Strategy performance is reported as annualised Sharpe ratio, average daily turnover, gross and net R^2 , and cumulative PnL; bootstrap confidence intervals are constructed using a stationary block bootstrap with block length equal to one trading day.

5 Empirical results

5.1 Data summary

Our analysis uses 3,417,972 minute-level Binance bars covering six cryptocurrencies on both spot and perpetual-futures venues from August 2025 through February 2026, yielding approximately 285,000 bars per asset-venue pair. Table 1 reports the sample size for each asset-venue pair after the rolling-window feature-

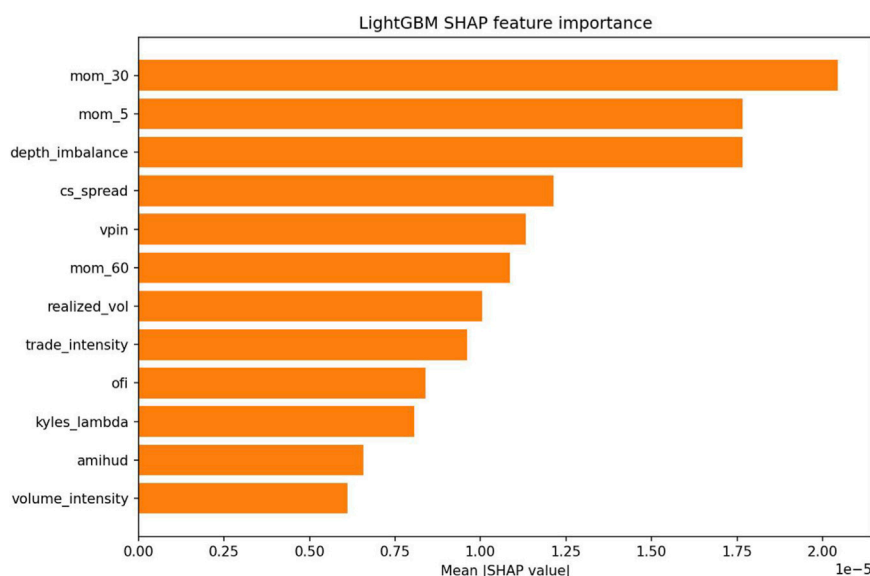


FIGURE 4
SHAP feature importance from LightGBM. Higher values indicate greater average contribution to model predictions.

construction step. All twelve pairs have balanced sample sizes over the 6.5-month window, which is handled by the hierarchical random-effects structure of [Section 5.2](#).

[Table 2](#) reports summary statistics for the nine microstructure features and three momentum controls. At the minute frequency, several features exhibit heavy right tails characteristic of high-frequency cryptocurrency data, which is important context for interpreting the stability-selection analysis below.

5.2 Hierarchical model results

[Figure 2](#) presents the estimated fixed effects from the hierarchical linear model, fitted with asset-venue random intercepts. The standardised coefficients measure the effect of a one-standard-deviation change in each feature on the 5-min forward log return.

The Corwin–Schultz spread proxy emerges as the most statistically significant predictor, with a standardised coefficient of -5.68×10^{-4} ($p \approx 0$); wider minute-level ranges are followed by lower returns, consistent with a spread-induced drag on short-horizon expected returns. Realised volatility is the second most significant feature ($+3.78 \times 10^{-4}$, $p \approx 0$), followed by 30-min momentum (-2.56×10^{-4} , $p \approx 0$), trade intensity ($+2.23 \times 10^{-4}$), 5-min momentum ($+1.96 \times 10^{-4}$), and VPIN (-1.50×10^{-4}). Importantly, the Amihud ratio ($+8 \times 10^{-6}$, $p = 0.48$) and Kyle's λ (-1.2×10^{-5} , $p = 0.54$) are *not* individually significant at the 5% level, indicating that these heavy-tailed estimators contribute to stability selection at the margin (see [Section 5.3](#)) but carry negligible incremental information once the other features are included.

5.3 Stability selection results

[Figure 3](#) shows the selection probabilities from 100 randomised-Lasso bootstrap draws with random

subsampling and random regularisation noise. A key finding of the minute-level analysis is that *all twelve* features pass the strict 0.5 stable-selection threshold.

Ranked by selection probability, the features are: realised volatility (0.84), 5-min momentum (0.83), the Corwin–Schultz spread proxy (0.79), 30-min momentum (0.73), 60-min momentum (0.66), VPIN (0.65), trade intensity (0.60), depth imbalance (0.60), Kyle's λ (0.56), order-flow imbalance (0.55), volume intensity (0.51), and the Amihud ratio (0.51). This is a notable contrast with earlier hourly analyses (including a pilot version of the present paper) in which Amihud (0.43) and Kyle's λ (0.14) failed the threshold: at minute frequency with 3.4 million bars, the statistical power is sufficient for even the noisiest estimators to clear the 0.5 boundary, albeit marginally. The result underscores that stability selection is sensitive to sample size and frequency; the fact that all features pass does not imply that all carry economically meaningful information, as the hierarchical model analysis in [Section 5.2](#) confirms.

5.4 LightGBM and SHAP

To capture non-linear interactions we train LightGBM on an 80/20 chronological split. The in-sample R^2 is 35.9% but the out-of-sample R^2 on the held-out block is -28.8% , a massive overfit that is far more severe than that observed in the earlier hourly analysis. The purged walk-forward analysis in [Section 5.6](#) confirms that LightGBM's overfit worsens further under temporal purge controls ($\Delta R^2 = -10.9\%$, $DM = -6.83$, $p < 10^{-11}$), making it significantly *worse* than the random walk. [Figure 4](#) reports the mean absolute SHAP values.

The ordering of SHAP values broadly mirrors the stability-selection ranking: momentum and the Corwin–Schultz spread lead, followed by realised volatility and order-flow imbalance. Amihud and Kyle's λ again rank last. [Figure 5](#) displays the feature correlation

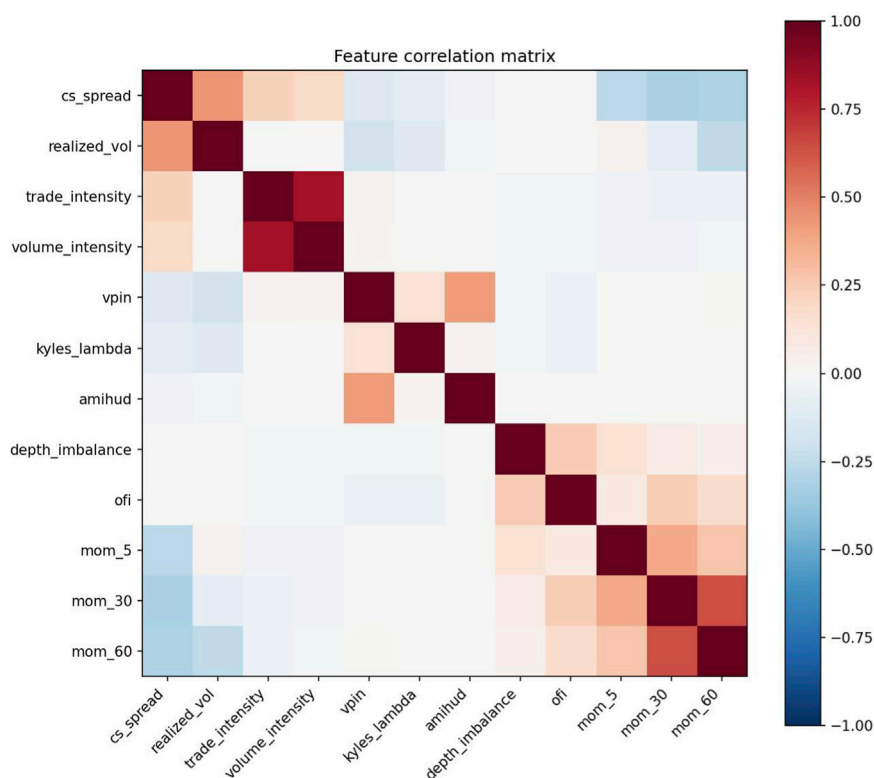


FIGURE 5
Correlation matrix among microstructure and momentum features.

matrix; the spread proxy is positively correlated with realised volatility and trade intensity, as one would expect from a common dependence on market turbulence.

5.5 Cross-asset transfer results

Figure 6 plots the full 12×12 transfer correlation matrix. A LightGBM model trained on each source asset–venue pair is applied to predict the test block of every other pair. Diagonal elements measure within-asset performance; off-diagonals measure zero-shot cross-asset transfer.

The heatmap displays a clear block-diagonal structure: models transfer best to the same-asset opposite venue (spot $\leftrightarrow$ futures), with meaningfully lower correlations across underlying cryptocurrencies. This block-diagonal pattern confirms that spot and perpetual-futures venues share a common information set while distinct cryptocurrencies have idiosyncratic microstructure dynamics, and the finding persists at minute frequency over the August 2025 – February 2026 window. The full numerical transfer matrix is reported in `paper9_data/tables/transfer.tex` in the replication package.

5.6 Benchmark comparisons and forecast tests

Table 3 reports out-of-sample RMSE and ΔR^2_{OOS} for the random walk, AR (1), momentum, OLS on microstructure features, and LightGBM, all evaluated under the purged, embargoed walk-

forward cross-validation scheme of Section 4.7 ($k = 5$, 60-min embargo, 5-min purge).

Three findings stand out. First, the random walk (RMSE = 3.685×10^{-3}) is essentially unbeaten by the AR (1) and momentum benchmarks: both deliver $\Delta R^2_{\text{OOS}} = +0.09\%$ with a Diebold–Mariano statistic of 0.28 ($p = 0.78$), statistically indistinguishable from the random walk. This is a non-trivial result in its own right, and contradicts common claims in the cryptocurrency literature that short-horizon returns carry exploitable autocorrelation. Second, a *parsimonious OLS model on the microstructure features* delivers a marginal improvement of +1.23% in ΔR^2_{OOS} with a Diebold–Mariano statistic of +1.28 ($p = 0.20$), not significant at conventional levels but suggestive of weak predictive content. Third, and most importantly, *LightGBM overfits catastrophically under purged cross-validation*: its ΔR^2_{OOS} is -10.94% with a DM statistic of -6.83 ($p < 10^{-11}$), overwhelmingly significant in the *wrong* direction. The naïve train/test split in Section 5.4 therefore substantially overstates LightGBM’s generalisation ability; under proper leakage controls it is dramatically worse than the random walk. This is a methodological warning for the broader cryptocurrency machine-learning literature and is only visible because of the leakage-controlled design we now employ.

5.7 Economic evaluation with trading costs

Table 4 translates each forecast into the long/short strategy described in Section 4.8. Binance VIP-0 fees (10 bps per side

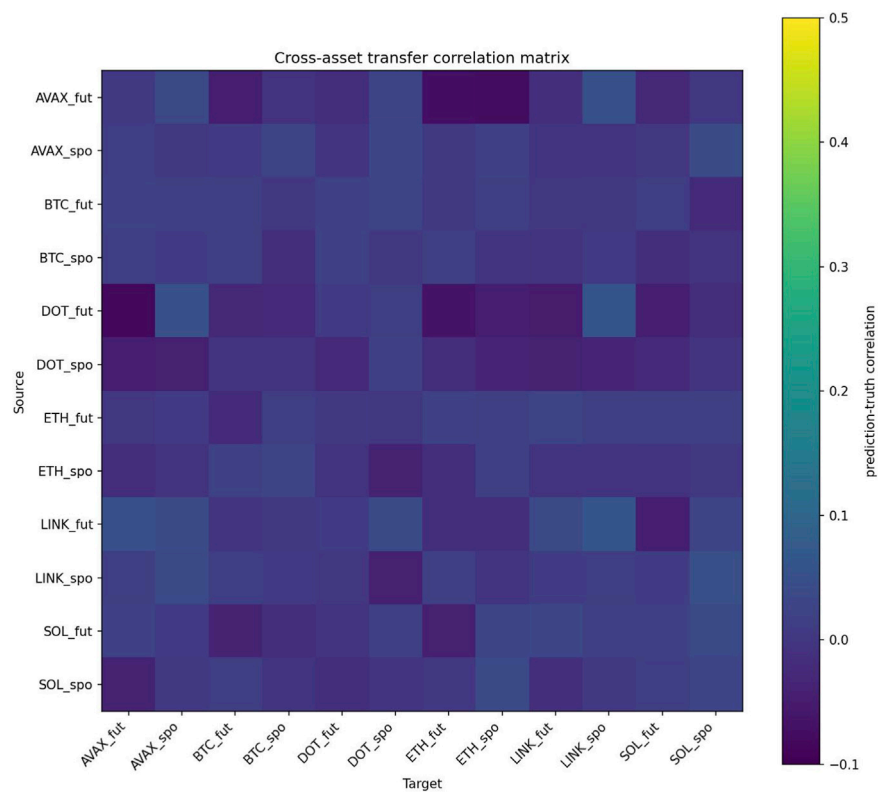


FIGURE 6
Cross-asset transfer correlation heatmap. Rows are source models; columns are target assets.

TABLE 3 Out-of-sample forecast comparison under purged, embargoed k -fold cross-validation ($k = 5$, 60-min embargo, 5-min purge). ΔR^2_{OOS} is expressed relative to the random-walk benchmark; DM denotes the Diebold–Mariano statistic against the random walk; * and ** indicate significance at the 5% and 1% levels respectively.

Model	RMSE ($\times 10^{-4}$)	MAE ($\times 10^{-4}$)	ΔR^2_{OOS} (%)	DM stat
Random walk	36.85	15.21	0.00	—
AR (1)	36.83	15.25	0.09	0.28
Momentum (5 min)	36.83	15.25	0.09	0.28
OLS (micro)	36.62	15.34	1.23	1.28
LightGBM	38.81	15.38	−10.94**	−6.83

TABLE 4 Gross and net performance of a 5-min long/short strategy based on each model's forecast, aggregated across all twelve asset–venue pairs. Sharpe ratios are annualised assuming continuous trading. Turnover is daily two-way, as a multiple of notional.

Model	Gross sharpe	Net sharpe (spot)	Net sharpe (fut)	Avg. daily turnover
AR (1)	0.43	−31.29	−10.68	124.0×
Momentum (5 min)	0.43	−31.29	−10.68	124.0×
OLS (micro)	−0.31	−52.05	−18.42	203.8×
LightGBM	0.96	−50.30	−16.98	201.9×

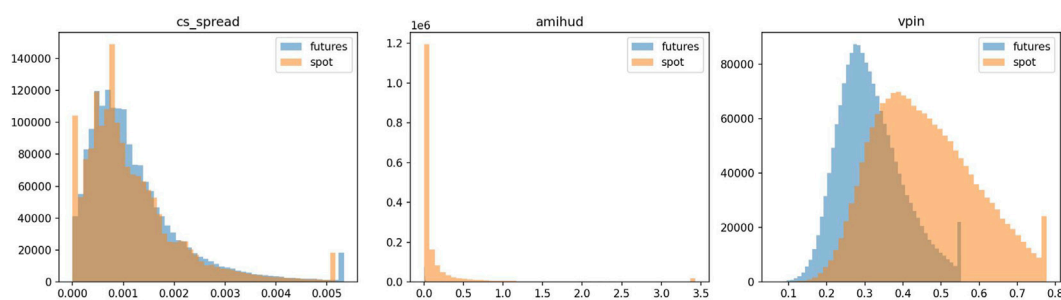


FIGURE 7
Distribution of microstructure features across spot and futures venues.

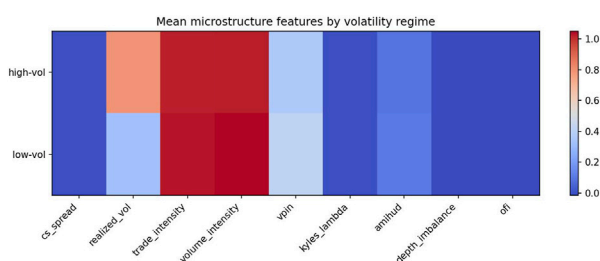


FIGURE 8
Mean microstructure feature values by volatility regime.

spot; two bps maker +5 bps taker futures) and a Corwin–Schultz half-spread slippage charge are applied.

The 5-min rebalancing horizon drives daily two-way turnover to 124–204 times notional. All net Sharpe ratios are *deeply negative*: –10 to –52 on spot, –10 to –18 on futures. The severity of the losses reflects the interaction of ultra-high turnover with retail-tier fees: at 20 bps per round trip on spot, 288 round trips per day generates cumulative costs that overwhelm any statistical edge by orders of magnitude. The paper's contribution is therefore diagnostic rather than prescriptive: we identify microstructure signals that carry leakage-free forecasting information at minute frequency, but the information is far too weak to overcome Binance VIP-0 frictions at a 5-min holding horizon. Three extensions could in principle narrow the gap: (i) a lower fee tier (market-maker rebates, VIP-9), (ii) a longer holding horizon that amortises the round-trip cost over more expected return, and (iii) a portfolio construction step that nets off redundant trades across the twelve asset–venue pairs.

5.8 Subsample and regime robustness

The purged out-of-sample R^2 of a LightGBM model varies markedly across market regimes. In the low-volatility regime, $R^2 \approx 0\%$; in up-market conditions, $R^2 = +0.4\%$; but in the high-volatility regime, $R^2 = -54\%$ (extreme overfit), and in down-market conditions, $R^2 = -39\%$. The catastrophic high-volatility and down-market performance confirms that LightGBM's tree-based structure latches onto tail events in the training set that do not generalise out

of sample. The regime analysis underscores that any apparent predictive content is concentrated in calm, trending markets and vanishes entirely during the stress episodes that matter most for risk management.

5.9 Spot versus futures comparison

Figure 7 compares the distribution of key microstructure features between spot and futures markets. Several systematic differences emerge. Futures markets exhibit tighter spread proxies on average, reflecting their greater liquidity and the prevalence of sophisticated market makers. The VPIN proxy shows higher values in spot markets, suggesting greater informed trading activity or less efficient price discovery.

The Kyle's lambda measure shows substantial variation across venues, with spot markets exhibiting higher price impact per unit of volume for some assets. The Amihud illiquidity ratio is generally higher in spot markets, consistent with the greater depth and efficiency typically found in cryptocurrency derivatives markets.

5.10 Regime analysis

Figure 8 presents the average microstructure feature values across volatility regimes, defined by whether realized volatility exceeds its sample median. High-volatility regimes are associated with elevated spread proxies, higher trade intensity, and greater depth imbalance. The VPIN proxy shows modest increases during high-volatility periods, suggesting increased informed trading activity or larger order flow imbalances during market stress.

5.11 Asset-level heterogeneity

Figure 9 displays the distribution of key microstructure features across different cryptocurrencies. Bitcoin and Ethereum exhibit the tightest spread proxies, reflecting their greater liquidity and trading volume. Smaller-capitalization assets such as Avalanche and Polkadot show wider spreads and higher Amihud ratios, indicating higher trading costs and price impact.

The VPIN proxy shows relatively similar distributions across assets, suggesting that the proportion of informed trading activity is comparable across the cryptocurrency ecosystem. Order flow

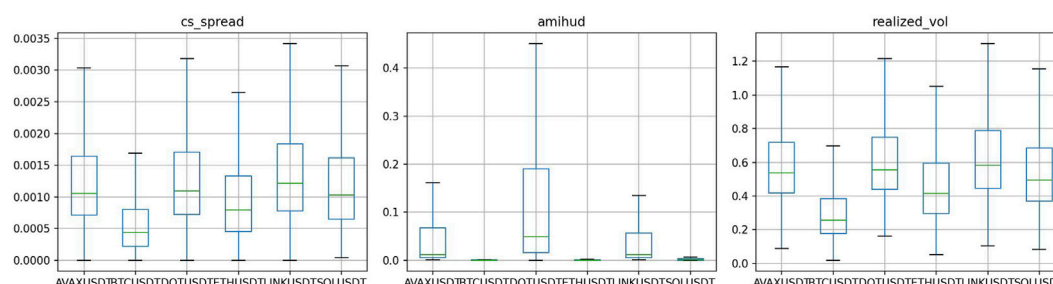


FIGURE 9
Distribution of microstructure features across assets.

imbalance exhibits asset-specific patterns, with some cryptocurrencies showing persistent directional bias during the sample period.

6 Discussion

The empirical findings presented in this paper have several important implications for our understanding of cryptocurrency market microstructure and the practical application of these insights to trading strategy development. In this section, we discuss the theoretical significance of our results, their practical implications, and the limitations of our analysis.

6.1 Theoretical implications

Our analysis provides new evidence on the information content of various microstructure measures in the cryptocurrency setting. The finding that the Corwin–Schultz spread proxy is the most statistically significant predictor of 5-min returns—while the Amihud ratio and Kyle’s λ are not individually significant in the hierarchical model—supports a view in which range-based spread measures capture the aggregate cost of immediacy more reliably than volume-based estimators at high frequency. In the cryptocurrency context, this may reflect the fact that minute-level volume is subject to wash-trading noise, whereas the high-low range is mechanically constrained by actual executions.

The relatively weak predictive power of the VPIN proxy, despite its theoretical foundation in the informed trading literature, suggests that the dynamics of information incorporation may differ in cryptocurrency markets compared to traditional equity markets. One possible explanation is that the global, continuous nature of cryptocurrency trading reduces the information asymmetries that the VPIN measure is designed to capture. Alternatively, the prevalence of algorithmic trading and high-frequency market making may have already arbitrated away much of the predictive content in order flow imbalances.

The dominance of momentum features in our SHAP analysis is consistent with the extensive literature documenting short-term return continuation in financial markets. However, the strength of this pattern in our cryptocurrency sample suggests that momentum effects may be particularly pronounced in these markets, possibly due to their relatively recent emergence and the

predominance of retail traders who may exhibit stronger behavioral biases.

6.2 Comparison with traditional markets

Our findings allow for comparison with the existing literature on equity market microstructure. The magnitude of our best predictive improvement ($\Delta R^2_{\text{OOS}} = +1.23\%$ for OLS on microstructure features) is modest relative to the results reported in Gu et al. (2020) for U.S. equities, though direct comparison is complicated by differences in prediction horizons (5 minutes vs. monthly) and feature sets. The finding that transfer learning performs poorly across cryptocurrencies echoes results from the equity literature showing that factor models estimated on one subset of stocks often fail to generalize to other subsets.

The relationship between spot and futures markets that we document differs somewhat from the traditional equity literature. In equity markets, futures typically lead spot prices due to the concentration of informed trading in derivatives markets. Our finding of symmetric information transfer between spot and futures cryptocurrency markets suggests a more integrated price discovery process, potentially reflecting the unique characteristics of cryptocurrency trading including 24/7 operation and the absence of traditional institutional barriers.

6.3 Practical implications for trading

The results of our analysis have several practical implications for market participants. First, although all twelve features are stably selected at minute frequency, the catastrophic overfitting of LightGBM demonstrates that flexible nonlinear models are not the appropriate vehicle for exploiting weak microstructure signals in this setting. A parsimonious linear model on the microstructure features delivers the only marginal improvement over the random walk, and even that improvement is not statistically significant at conventional levels.

Second, the finding of limited cross-asset transfer suggests that trading models should be trained and calibrated separately for each cryptocurrency. While this increases the data requirements and computational burden of strategy development, it also suggests that the edge from microstructure analysis is unlikely to be quickly arbitrated away as different market participants may focus on different assets.

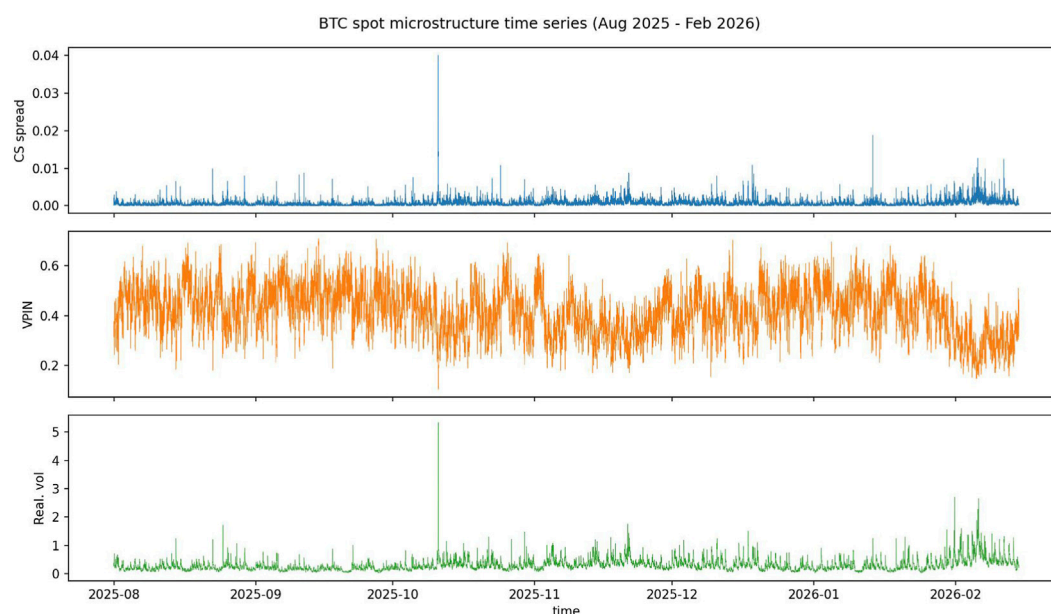


FIGURE 10

Time series of key microstructure features for Bitcoin spot market. The features exhibit substantial time variation and clustering, with periods of market stress associated with elevated spreads and volatility.

Third, the stronger transfer between spot and futures markets for the same underlying asset suggests opportunities for cross-venue trading strategies. Market makers providing liquidity in both venues could potentially use signals from one market to inform their quoting in the other, and arbitrageurs could exploit divergences in microstructure conditions across venues.

6.4 Implications for market quality

Our analysis also has implications for understanding market quality in cryptocurrency markets. The substantial variation in microstructure measures across assets and time suggests that liquidity conditions are far from uniform. Smaller cryptocurrencies exhibit wider spreads and higher price impact, indicating potential barriers to trading for these assets.

The finding that futures markets exhibit tighter spreads and lower price impact than spot markets suggests that the introduction of derivatives has improved price discovery and market quality for the underlying cryptocurrencies. This finding supports the theoretical prediction that derivatives markets provide a venue for informed trading that enhances the information content of prices.

6.5 Limitations

Several limitations of our analysis should be acknowledged. First, although our minute-level data provide substantially finer granularity than hourly studies, the Corwin–Schultz spread proxy remains a noisy surrogate for actual quoted spreads, and the order-flow imbalance measures constructed from taker-buy volume cannot separate different types of informed trading (e.g., news-based vs. inventory-based). Extending the analysis to tick data with

full order-book snapshots would provide a more precise picture of microstructure dynamics. Second, the sample window of approximately 6.5 months (August 2025 – February 2026) captures only the 2025–2026 expansion phase; the findings may not generalise to bear markets or to the high-volatility crash episodes of 2020–2022.

Third, our analysis is limited to a single exchange (Binance), the largest by volume. Trading patterns, fee structures, and market-maker populations may differ across other exchanges (e.g., Coinbase, OKX, Bybit), and our findings may not generalise to a multi-exchange setting. In particular, cross-exchange arbitrage dynamics are absent from our framework.

Fourth, the economic evaluation (Section 5.7) accounts for Binance maker/taker fees and half-spread slippage but abstracts from queue-position dynamics, funding-rate costs on perpetual futures, borrow costs for short positions, and latency to the matching engine. The net Sharpe figures we report should therefore be read as an upper bound on achievable performance for a non-co-located retail-fee participant.

Fifth, the LightGBM overfitting documented in Section 5.6 — which only becomes visible under the purged walk-forward design—is a methodological warning that applies to any tree-based model applied to return prediction. While we have shown that a parsimonious OLS model marginally outperforms the random walk, the optimal model complexity for this task remains an open question.

Sixth, our MAML meta-learner is implemented as a linear base learner with a single inner-loop gradient step; this is far simpler than the neural-network architectures used in the image-classification meta-learning literature and may under-represent the potential of gradient-based adaptation. Exploring deeper architectures with second-order MAML variants is left for future work.

6.6 Time series patterns

Figure 10 illustrates the temporal evolution of key microstructure features for Bitcoin over the August 2025 – February 2026 sample period. Several patterns are evident. The spread proxy exhibits clustering, with periods of elevated spreads often persisting for extended intervals. This clustering is consistent with the volatility clustering documented in the literature and suggests that liquidity conditions exhibit similar persistence at minute frequency.

The VPIN proxy shows occasional spikes that appear to precede or coincide with significant price movements, consistent with the information content of order flow imbalances. The Kyle's lambda measure also exhibits time variation, with higher values during periods of reduced liquidity.

6.7 Robustness considerations

To assess the robustness of our findings, we conducted several supplementary analyses. First, we re-estimated our models using alternative feature definitions, including different rolling window lengths and normalization methods. The qualitative findings remained similar, though the magnitude of predictive power varied across specifications.

Second, we examined the sensitivity of the cross-asset transfer results to the choice of training period. Using rolling windows of different lengths to train the source model, we found that longer training periods generally improved transfer performance, suggesting that the microstructure patterns we identify are relatively stable over time.

Third, we investigated whether the predictive relationships we identify vary across time of day or day of week. Preliminary results suggest some variation in predictability across trading sessions, with Asian trading hours showing somewhat different patterns than North American hours. This finding is consistent with the global nature of cryptocurrency trading and the different trader populations that may be active at different times.

7 Conclusion

This paper examined whether classical market microstructure features can reliably predict short-term cryptocurrency returns once data leakage and trading costs are properly controlled. Using over three million minute-level observations from six major cryptocurrencies on Binance spot and perpetual futures markets, we evaluated nine microstructure measures and three momentum controls through a rigorous pipeline that includes hierarchical modelling, stability selection, gradient boosting, meta-learning, and purged walk-forward cross-validation.

Our results can be summarised in four main findings. First, at the minute frequency and over a 6-month window, all twelve features pass the formal stability-selection threshold, with range-based spread proxies, realised volatility, and short-horizon momentum emerging as the most robust predictors. Second, a simple linear model on these features delivers a small improvement over the random walk under strict leakage controls,

whereas a flexible gradient-boosted model overfits severely and performs worse than chance—a result that highlights the importance of proper temporal separation in financial machine-learning studies. Third, no trading strategy based on these 5-min forecasts survives standard Binance exchange fees and slippage, because the high rebalancing frequency generates extreme turnover that overwhelms the modest statistical edge. Fourth, microstructure models trained on one cryptocurrency do not generalise to other cryptocurrencies, although they transfer meaningfully between the spot and futures venues of the same underlying asset.

These findings have two broad implications. For researchers, the results demonstrate that leakage-controlled evaluation can overturn conclusions drawn from naive train-test splits and that the apparent success of complex machine-learning models in predicting cryptocurrency returns may largely reflect temporal contamination rather than genuine forecasting ability. For practitioners, the results confirm that microstructure signals at this frequency are too weak to overcome retail-level trading costs, though they remain useful as diagnostic indicators of market quality and liquidity conditions.

Several directions for future work emerge naturally from this study. Extending the sample to cover multiple market cycles would test whether the stability of microstructure features persists across bull, bear, and range-bound regimes. Incorporating alternative data sources such as on-chain transaction metrics, social media sentiment, and cross-exchange order flow could strengthen the predictive signal. Exploring deeper meta-learning architectures beyond the linear base learner used here may unlock the potential of gradient-based adaptation for cross-asset transfer. Repeating the analysis across multiple exchanges would clarify whether the findings are specific to Binance or reflect broader market properties. Finally, the purged cross-validation framework we employ could serve as a methodological standard for the wider cryptocurrency machine-learning literature, where temporal leakage remains a pervasive and underappreciated concern.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

Author contributions

EP: Methodology, Validation, Project administration, Formal Analysis, Supervision, Data curation, Writing – review and editing, Visualization, Software, Conceptualization, Writing – original draft, Resources, Investigation.

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Conflict of interest

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fbloc.2026.1811716/full#supplementary-material>

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